A SECTION VANISHING ON AN OPEN CONTAINING THE WEAKLY ASSOCIATED POINTS VANISHES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Weakly associated points behave for short exact sequences. Lemma lemma-ses-weakly-ass, used at proof lines 663-665. Gives WeakAss(F') contained in WeakAss(F).
- Weakly associated points lie in the support. Lemma lemma-weakly-ass-support, used at proof lines 665-666. Gives WeakAss(F') contained in Supp(F').
- A quasi-coherent module with no weakly associated points is zero. Lemma lemma-weakly-ass-zero, used at proof lines 667-668. Concludes F' = 0, hence s = 0.

Source reference. Original lemma 05AN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6856–6856. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05AM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6855–6855. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05AP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6857–6857. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. If $U \subset X$ is open and $\mathrm{WeakAss}(\mathcal{F}) \subset U$, then $\Gamma(X, \mathcal{F}) \to \Gamma(U, \mathcal{F})$ is injective.*

Proof. Let $s \in \Gamma(X, \mathcal{F})$ be a section which restricts to zero on U. Let $\mathcal{F}' \subset \mathcal{F}$ be the image of the map $\mathcal{O}_X \to \mathcal{F}$ defined by s. Then $\mathrm{Supp}(\mathcal{F}') \cap U = \emptyset$. On the other hand, $\mathrm{WeakAss}(\mathcal{F}') \subset \mathrm{WeakAss}(\mathcal{F})$ by Lemma 05AN. Since also $\mathrm{WeakAss}(\mathcal{F}') \subset$

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$\mathrm{Supp}(\mathcal{F}')$ (Lemma 05AM) we conclude $\mathrm{WeakAss}(\mathcal{F}') = \emptyset$. Hence $\mathcal{F}' = 0$ by Lemma 05AP. $\square$